Chapter 5: Multiple Linear Regression

Foundations, Least Squares, Geometry, and the Overall F-Test

Study notes with extensions for later research
Version 1.0 | October 14, 2026

Scope. These notes consolidate the concepts discussed so far, starting with the model and design matrix and ending at Section 5.6.1. The organization below is pedagogical; Sections 5.1–5.5 are not a verified transcription of the lecture's subsection headings. Section 5.6.1 follows the supplied screenshots. Research extensions are explicitly identified. Throughout, $p = m + 1$ counts the intercept and m predictor columns.

Contents

5 Multiple Linear Regression **2**

5.1 The model, variables, and notation 2

5.2 The design matrix: rows, columns, and dimensions 2

5.3 Sample size, rank, and identifiability 3

5.4 Least squares: gradient, normal equations, and Hessian 4

5.5 Projection, orthogonality, centering, and inference foundations 5

5.6 Testing the fit: the basic ANOVA table 6

5.6.1 The overall F-test 6

Research extensions **9**

5 Multiple Linear Regression

5.1 The model, variables, and notation

For observation i,

$$Y_i = \beta_0 + \sum_{j=1}^{m} \beta_j x_{ij} + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Here Y_i is a response, x_{ij} is predictor j for observation i, β_0 is the intercept, and ε_i is the unobserved error. A coefficient describes a change in the modeled mean while the other predictors are held fixed; this is not automatically a causal effect.

Linear means linear in the unknown coefficients. The model

$$Y_i = \beta_0 + \beta_1 x_{i1}^2 + \beta_2 \log x_{i2} + \varepsilon_i$$

is linear in its parameters (with $x_{i2} > 0$). In contrast,

$$Y_i = \beta_0 + \beta_1 e^{\beta_2 x_{i1}} + \beta_3 x_{i2} + \varepsilon_i$$

is nonlinear in its parameters because the unknown β_2 is inside the exponential.

Predictor versus covariate. Both can be numerical or categorical. "Covariate" often emphasizes adjustment; traditional ANCOVA terminology reserves it for a numerical variable and calls a categorical variable a factor.

For a quantitative response, the following are conventional descriptions. Let T_i indicate treatment and S_i indicate smoking, each coded 0 or 1.

Explanatory types	Common model name	Example mean $E(Y_i \mid X)$
Numerical + numerical	Multiple linear regression	$\beta_0 + \beta_1 \text{Exercise}_i + \beta_2 \text{Age}_i$
Numerical + categorical	ANCOVA	$\beta_0 + \beta_1 \text{Exercise}_i + \beta_2 S_i$
Categorical + numerical	ANCOVA	$\beta_0 + \beta_1 T_i + \beta_2 \text{Age}_i$
Categorical + categorical	Additive two-way ANOVA	$\beta_0 + \beta_1 T_i + \beta_2 S_i$

All belong to the general linear model framework. Add $\beta_3 T_i S_i$ if the treatment-group contrast is allowed to vary by smoking group. A factor with k categories usually uses $k - 1$ indicators when an intercept is included. The number m counts design columns, not necessarily distinct scientific variables.

5.2 The design matrix: rows, columns, and dimensions

$$Y = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1m} \\ 1 & x_{21} & \cdots & x_{2m} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nm} \end{pmatrix}.$$

Object	Meaning	Dimension
$Y, \boldsymbol{\varepsilon}$	Response and error vectors	$n \times 1$
X	Full design matrix	$n \times p$
$\boldsymbol{\beta} = (\beta_0, \ldots, \beta_m)^T$	Coefficient vector	$p \times 1$
$\mathbf{1}_n$	Intercept column of ones	$n \times 1$
X_s	Design excluding the intercept	$n \times m$

Avoid a notation collision. Lecture notes sometimes use $\mathbf{x}_i$ for an observation vector and $\mathbf{x}_j$ for a predictor column. They are different objects. Here we use

$$r_i = \begin{pmatrix} 1 \\ x_{i1} \\ \vdots \\ x_{im} \end{pmatrix} \in \mathbb{R}^p, \qquad c_j = \begin{pmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{pmatrix} \in \mathbb{R}^n.$$

Then

$$X = \begin{pmatrix} r_1^T \\ \vdots \\ r_n^T \end{pmatrix} = (\mathbf{1}_n \; c_1 \; \cdots \; c_m) = (\mathbf{1}_n \; X_s).$$

The transpose makes an observation column vector into a row. It does not, by itself, indicate whether a vector represents an observation or a predictor.

Example. For ages 20, 30, 40 and weights 60, 70, 80,

$$X = \begin{pmatrix} 1 & 20 & 60 \\ 1 & 30 & 70 \\ 1 & 40 & 80 \end{pmatrix}, \quad r_1^T = (1, 20, 60), \quad c_1 = (20, 30, 40)^T, \quad c_2 = (60, 70, 80)^T.$$

The scalar $x_{12} = 60$ is person 1's weight. Notice that this particular illustrative matrix is rank deficient: weight equals age plus 40, so it should not be used as a full-rank estimation example.

5.3 Sample size, rank, and identifiability

Full column rank means $\text{rank}(X) = p$: no design column is an exact linear combination of the others.

Setting	Terminology	OLS consequence
$n \gg p$	Classical, low dimensional	Unique coefficients if full column rank holds.
$n > p$	Overdetermined	Full column rank is possible; residual df is $n - p$ if full rank.
$n = p$	Square; saturated if full rank	Full-rank fit interpolates Y; no residual df.
$n < p$	Underdetermined; often high dimensional	Rank is less than p; coefficients are not unique.

Having $n > p$ does not guarantee rank p. For instance,

$$X = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 4 \\ 1 & 3 & 6 \\ 1 & 4 & 8 \end{pmatrix}$$

has $n = 4 > p = 3$ but its last column is twice its second. Only $\beta_1 + 2\beta_2$ is identified, not the two slopes separately.

Research extension. Dimensionality is also about the ratio p/n and how it changes with sample size. Problems with p close to n can show high-dimensional behavior even with $p < n$. This statistical issue is distinct from algebraic invertibility.

5.4 Least squares: gradient, normal equations, and Hessian

Step 1: minimize squared residuals

$$S(\boldsymbol{\beta}) = \|Y - X\boldsymbol{\beta}\|^2 = Y^T Y - 2\boldsymbol{\beta}^T X^T Y + \boldsymbol{\beta}^T X^T X \boldsymbol{\beta}.$$

The two cross terms combine because $Y^T X\boldsymbol{\beta} = \boldsymbol{\beta}^T X^T Y$ is a scalar.

Step 2: differentiate

For fixed a and symmetric A,

$$\nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^T a) = a, \qquad \nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^T A\boldsymbol{\beta}) = 2A\boldsymbol{\beta}.$$

Therefore,

$$\nabla S(\boldsymbol{\beta}) = -2X^T Y + 2X^T X\boldsymbol{\beta} \in \mathbb{R}^{p\times 1}.$$

Step 3: solve the normal equations

Setting the gradient to zero gives

$$\boxed{X^T X\widehat{\boldsymbol{\beta}} = X^T Y.}$$

If $\operatorname{rank}(X) = p$,

$$\boxed{\widehat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T Y.}$$

Although X is rectangular, $X^T X$ is $p \times p$ and square.

Step 4: verify the minimum using the Hessian

$$\boxed{\nabla^2 S = 2X^T X.}$$

For every $v \in \mathbb{R}^p$,

$$v^T \nabla^2 S\, v = 2\|Xv\|^2 \geq 0.$$

Thus S is convex. If X has full column rank, the expression is strictly positive for every $v \neq 0$, so the Hessian is positive definite and the minimizer is unique.

If rank is deficient, there is a nonzero v with $Xv = 0$. Then

$$X(\widehat{\boldsymbol{\beta}} + tv) = X\widehat{\boldsymbol{\beta}} \quad \text{for all } t \in \mathbb{R}.$$

There are infinitely many minimizing coefficients, but a unique fitted vector. A useful global verification is

$$S(\widehat{\boldsymbol{\beta}} + d) = S(\widehat{\boldsymbol{\beta}}) + \|Xd\|^2,$$

because the normal equations make the cross term zero.

Worked example

Let

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad Y = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad X^T X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad X^T Y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

The normal equations are

$$3\widehat{\beta}_0 + 3\widehat{\beta}_1 = 5, \qquad 3\widehat{\beta}_0 + 5\widehat{\beta}_1 = 6.$$

Subtracting gives $2\widehat{\beta}_1 = 1$, hence

$$\widehat{\beta}_1 = \tfrac{1}{2}, \qquad \widehat{\beta}_0 = \tfrac{7}{6}.$$

The Hessian is $\begin{pmatrix} 6 & 6 \\ 6 & 10 \end{pmatrix}$ with leading principal minors $6 > 0$ and $24 > 0$, so the minimum is unique. The fitted values and residuals are

$$\widehat{Y} = \begin{pmatrix} 7/6 \\ 5/3 \\ 13/6 \end{pmatrix}, \qquad e = \begin{pmatrix} -1/6 \\ 1/3 \\ -1/6 \end{pmatrix}, \qquad SSE = \tfrac{1}{6}.$$

Check: $\mathbf{1}^T e = 0$ and $(0, 1, 2)e = 0$.

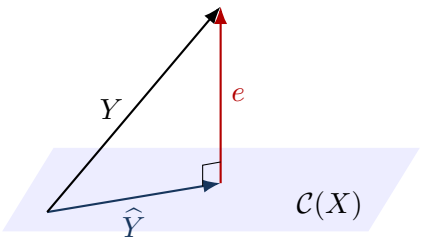
5.5 Projection, orthogonality, centering, and inference foundations

Geometry of the fit

The column space $\mathcal{C}(X) = \{Xb : b \in \mathbb{R}^p\}$ contains every possible model prediction. OLS projects Y onto this space:

$$Y = \widehat{Y} + e, \qquad \widehat{Y} \in \mathcal{C}(X), \qquad X^T e = 0.$$

Residual orthogonality follows from optimization; the predictors themselves need not be mutually orthogonal.



Schematic: the residual is perpendicular to the entire fitted space.

When full rank holds, define the hat matrix

$$H = X(X^T X)^{-1} X^T, \qquad \widehat{Y} = HY, \qquad e = (I - H)Y.$$

Its essential properties are

$$H^T = H, \quad H^2 = H, \quad \operatorname{rank}(H) = \operatorname{tr}(H) = p, \quad H(I - H) = 0.$$

A projection applied twice equals the same projection. The Hessian $2X^T X$ is $p \times p$; the hat matrix H is $n \times n$. They are different matrices.

Centering and the intercept-only projection

$$H_1 = \frac{1}{n}\mathbf{1}\mathbf{1}^T, \qquad H_1 Y = \bar{Y}\mathbf{1}, \qquad C = I - H_1.$$

Multiplication by C subtracts the mean. In particular,

$$X_c = CX_s = X_s - \mathbf{1}\bar{x}^T, \qquad \bar{x}^T = (\bar{x}_1, \ldots, \bar{x}_m).$$

Every column of X_c sums to zero. This makes centered predictor columns orthogonal to the intercept column, but does not generally make them orthogonal to each other. With full rank, the slopes and intercept can be recovered as

$$\widehat{\boldsymbol{\beta}}_s = (X_c^T X_c)^{-1} X_c^T CY, \qquad \widehat{\beta}_0 = \bar{Y} - \bar{x}^T \widehat{\boldsymbol{\beta}}_s.$$

Statistical assumptions

Algebraic least squares does not require normality. For the following classical mean and variance results, condition on a fixed X and assume

$$E(\boldsymbol{\varepsilon} \mid X) = 0, \qquad \mathrm{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I.$$

Then

$$E(\widehat{\boldsymbol{\beta}} \mid X) = \boldsymbol{\beta}, \qquad \mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid X) = \sigma^2 (X^T X)^{-1}.$$

If $n > p$,

$$s^2 = \frac{SSE}{n-p}, \qquad E(s^2 \mid X) = \sigma^2.$$

Crucial distinction: $X^T e = 0$ is a sample fitting identity; $E(\boldsymbol{\varepsilon} \mid X) = 0$ is an assumption about the data-generating process. The former does not prove the latter.

5.6 Testing the fit: the basic ANOVA table

5.6.1 The overall F-test

Question. Do the predictors jointly improve on an intercept-only model?

$$H_0 : \beta_1 = \cdots = \beta_m = 0, \qquad H_1 : \text{at least one slope is nonzero.}$$

The intercept is not tested. Under H_0, the modeled response mean is constant.

Sums of squares. For a full-rank OLS model with an intercept,

$$SST = \sum_i (Y_i - \bar{Y})^2 = Y^T (I - H_1) Y,$$
$$SSR = \sum_i (\widehat{Y}_i - \bar{Y})^2 = Y^T (H - H_1) Y,$$
$$SSE = \sum_i (Y_i - \widehat{Y}_i)^2 = Y^T (I - H) Y.$$

The decomposition

$$Y - \bar{Y}\mathbf{1} = (\widehat{Y} - \bar{Y}\mathbf{1}) + e$$

is orthogonal, so the Pythagorean theorem gives

$$\boxed{SST = SSR + SSE.}$$

Thus $SSR = SST - SSE$ is the reduction in squared error relative to predicting the sample mean for everyone.

Mean squares and intuition.

$$MSR = \frac{SSR}{m}, \qquad MSE = \frac{SSE}{n-m-1}, \qquad \boxed{F = \frac{MSR}{MSE}.}$$

Under the stated mean and variance assumptions,

$$E(MSE) = \sigma^2, \qquad E(MSR) = \sigma^2 + \frac{\boldsymbol{\beta}_s^T X_c^T X_c \boldsymbol{\beta}_s}{m}.$$

The extra term is nonnegative because

$$\boldsymbol{\beta}_s^T X_c^T X_c \boldsymbol{\beta}_s = \|X_c \boldsymbol{\beta}_s\|^2 = \sum_i \left[\sum_j \beta_j (x_{ij} - \bar{x}_j) \right]^2.$$

It measures variation in the true modeled means around their average. Under H_0 both mean squares estimate σ^2. Under H_1, the regression mean square contains additional signal. Large F is therefore evidence against H_0. This does not assert $E(F) = 1$ under the null.

Why an F distribution? For the exact finite-sample distribution assume $\boldsymbol{\varepsilon} \sim N_n(0, \sigma^2 I)$. Under H_0,

$$SSR/\sigma^2 \sim \chi^2_m, \qquad SSE/\sigma^2 \sim \chi^2_{n-p}.$$

These quadratic forms are independent because the corresponding projections are orthogonal:

$$(H - H_1)(I - H) = 0.$$

Consequently,

$$\boxed{\frac{SSR/m}{SSE/(n-m-1)} \sim F_{m,n-m-1} \quad \text{under } H_0.}$$

The unknown σ^2 cancels in the ratio.

Quadratic-form result behind the test. If $Z \sim N_n(\mu, \sigma^2 I)$ and P is symmetric and idempotent with rank r, then

$$\frac{Z^T P Z}{\sigma^2} \sim \chi^2_r(\lambda), \qquad \lambda = \frac{\mu^T P \mu}{\sigma^2}.$$

Here the noncentrality parameter uses the standard convention without a factor $1/2$. If $\lambda = 0$, the chi-square is central. For symmetric projection matrices P, Q, the condition $PQ = 0$ gives independence of the corresponding normal quadratic forms. Normality is essential to this independence argument.

ANOVA and decision.

Source	df	Sum of squares	Mean square	Test statistic
Regression	m	SSR	SSR/m	MSR/MSE
Residual	$n-m-1$	SSE	$SSE/(n-m-1)$	
Total	$n-1$	SST		

At level α, reject for $F_{\text{obs}} > F_{\alpha;m,n-m-1}$, where F_α denotes an upper-tail critical value. Equivalently, reject if the upper-tail p-value is less than α.

Example. With $n = 10$, $m = 2$, $SSR = 60$, and $SSE = 14$,

$$MSR = 30, \qquad MSE = 2, \qquad F = 15,$$

compared against $F_{2,7}$ under the null. Rejection supports at least one nonzero slope, not that every predictor is important or that the relationship is causal. Failure to reject is not proof that all slopes are zero.

Power under the alternative. Under a fixed nonzero slope vector,

$$F \sim F_{m,n-p}(\lambda), \qquad \lambda = \frac{\boldsymbol{\beta}_s^T X_c^T X_c \boldsymbol{\beta}_s}{\sigma^2}.$$

Larger signal relative to error increases power. Under a stable, nondegenerate design with fixed m and fixed nonzero slopes, $n^{-1}X_c^T X_c \to Q$ positive definite implies λ grows proportionally to n and power tends to one. A mere upper bound $X_c^T X_c = O(n)$ alone does not establish this conclusion.

Research extensions: using the foundations responsibly

Purpose. The following are conceptual extensions, not a complete analysis protocol or a substitute for design-specific methods.

Diagnose the actual problem

Issue	Meaning	Response
Nonorthogonal columns	Usual correlations among predictors	OLS is valid algebraically if full rank.
Exact dependence	Coefficients are not separately identified	Remove redundant encoding or reparameterize; identify estimable combinations.
Near dependence	Some coefficient directions have little information	Assess uncertainty and sensitivity; consider ridge or better data.
$p > n$	Too many coefficient directions for unique OLS	Add explicit structure; evaluate prediction and uncertainty appropriately.
Misspecified mean	Chosen column space omits relevant structure	Use scientifically motivated transformations, interactions, or splines.
Unequal error variance	Classical covariance formula may fail	Consider appropriate robust covariance or justified weighted least squares.
Dependent observations	$\sigma^2 I$ is inappropriate	Use methods reflecting clustering, repeated measures, time, or sampling design.

Numerical computation and near dependence

The inverse formula is useful for theory. In computation, solve least squares using QR decomposition or SVD rather than explicitly inverting $X^T X$. If $X = QR$ has full column rank and $Q^T Q = I$, solve

$$R\widehat{\boldsymbol{\beta}} = Q^T Y, \qquad \widehat{Y} = QQ^T Y.$$

Orthogonal computational directions can be constructed even when the original predictors are correlated. SVD also reveals small singular values and rank deficiency. Forming $X^T X$ squares the 2-norm condition number in the full-rank case.

VIFs summarize inflation associated with linear relationships among slope columns: $VIF_j = 1/(1-R_j^2)$, where R_j^2 comes from regressing predictor j on the other predictors with an intercept. There is no universal VIF cutoff that determines which scientific variable should be removed. Centering helps interpret intercepts and some polynomial terms, but does not remove ordinary correlations between the original numerical predictors.

Pseudoinverse and regularization

A Moore–Penrose pseudoinverse gives $\widehat{\boldsymbol{\beta}} = X^+ Y$, the minimum Euclidean norm OLS solution. It selects one coefficient vector without making redundant parameters identifiable. In a rank-deficient model, $\widehat{Y} = XX^+ Y$ remains the unique orthogonal projection.

Ridge minimizes

$$\|Y - X\boldsymbol{\beta}\|^2 + \lambda \boldsymbol{\beta}^T D \boldsymbol{\beta}, \quad D = \mathrm{diag}(0, 1, \ldots, 1), \quad \lambda > 0,$$

leaving the intercept unpenalized. Its solution is

$$\widehat{\boldsymbol{\beta}}_\lambda = (X^T X + \lambda D)^{-1} X^T Y.$$

With a usual nonempty intercept column and all slopes penalized, this matrix is positive definite. The first-order condition is

$$X^T e_\lambda = \lambda D \widehat{\boldsymbol{\beta}}_\lambda,$$

so residuals are generally not orthogonal to all predictors, although they still sum to zero when the intercept is unpenalized. The ridge smoother is generally not an idempotent orthogonal projection.

LASSO replaces the squared slope penalty by $\lambda \sum_{j=1}^{m} |\beta_j|$ and can select zero coefficients. Its coefficient solution need not be unique without additional conditions. Tune penalties using a validation strategy suited to the data; standardize predictors within training folds when appropriate, and avoid information leakage.

Separate prediction, interpretation, and causality

- **Prediction:** prioritize performance on new data; use suitable cross-validation and assess calibration and errors relevant to the application.
- **Interpretation:** specify the target coefficient or contrast, justify adjustment variables, and report uncertainty and sensitivity to modeling choices.
- **Causality:** define the intervention and estimand; justify identification using the study design and causal assumptions. Neither the overall F-test nor $X^T e = 0$ rules out confounding.

Do not automatically remove an important confounder to reduce VIF. Do not treat post-selection ordinary standard errors as automatically valid after a data-driven search. A complex survey requires methods reflecting weights, strata, and clusters; ordinary iid formulas are not automatically appropriate.

A compact memory guide

Concept	Question answered
Column space	What response patterns can the model express?
Projection	Which allowed prediction is closest to the observed Y?
Residual orthogonality	Why can no coefficient adjustment further reduce squared error?
Full column rank	Are the coefficients representing that fit unique?
Positive-definite Hessian	Is the least-squares objective strictly convex?
Residual degrees of freedom	How many dimensions remain for estimating error variance?
Overall F-test	Is the joint improvement beyond an intercept-only model statistically convincing?

Final distinction. Unique estimation, stable estimation, correct specification, and causal identification are separate properties. A model may satisfy one and fail another.